

VAIBHAV TUKARAM GAIKWAD

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SKILLS

Programming Technologies	C, C++, C#, x86 GNU Assembly, Java, JavaScript
Frameworks/Tools	STL, Multithreading, Socket Programming, WebSocket, CUDA
Graphics APIs	Qt, .NET (WPF), MySQL
Platforms	OpenGL, OpenGL ES, WebGL, DirectX 11
	Windows, Linux, UNIX

EXPERIENCE

Software Engineer Standard Units Supply India Pvt. Ltd.	Jan 2025 - Present <i>Pune, MH</i>
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- Performing R&D on real-time **C++ EtherCAT services** for industrial machine communication with **~4 ms cycle times**.
- Developing C# (.NET) applications for configuring, running and monitoring factory automation actuators.

C++ Developer TechZest Global Solutions	June 2024 - Jan 2025 <i>Pune, MH</i>
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- Diagnosed and resolved a critical latency bottleneck in HTF software, reducing round-trip latency from **450ms to 180ms with a 60% improvement** via optimized TCP/UDP socket communication.
- Applied multithreading patterns (**thread pools, mutexes, condition variables**) to maximize system throughput, improving processing speed by **~40%**.
- Refactored data management modules using optimized **lock-free queues**, reducing memory latency by **~35%**.

Software Developer Intern (C++) Avhan Technologies Pvt. Ltd.	Feb 2024 - June 2024 <i>Mumbai, MH</i>
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- Enhanced real-time data processing components for telecom-sector applications handling **high-frequency data streams** with minimal overhead.
- Implemented efficient data structures to process **10,000+ events/sec** with consistent low-latency performance.

PROJECTS

Bingo: A high-performance C++ WebSocket chat server supporting **500+ concurrent sessions** with thread-safe message routing and MySQL persistence. ([GitHub](#)) 2024

Customized Virtual File System: A Linux-like user-space file system built in C with custom inode/block structures simulating a **800MB standalone storage device**. ([GitHub](#)) 2023

CUDA VFX: A GPU-accelerated real-time video effects engine built with CUDA, achieving **~20x** speedup over CPU for per-pixel effects like grayscale, sepia, and hue shift on 1080p frames. ([GitHub](#)) 2025

COURSEWORK

C & C++ Programming | Data Structures & Algorithms | UNIX System Design | GNU x86 Assembly | Real-Time Rendering | Heterogeneous Parallel Programming

EDUCATION

Bharati Vidyapeeth, Pune Astromedicomp	Bachelor of Computer Applications Real-Time Rendering - Batch 6	Expected 2027 2024 - 2026
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